COMPUTER SECURITY

DCS-01-0020/2025

TITUS VUNDI WAMBUA

ASSIGNMENT ONE.

1. ***Analyze the key components of DES operation.***

1. Block and Key Dimensions : DES is a symmetric block cipher that processes data in 64-bit plaintext blocks using an effective 56-bit key .

2. Feistel Architecture: The algorithm processes data through 16 identical rounds of a Feistel network. In each round, the 64-bit block is split into Left and Right 32-bit halves; only the Right half is processed by the encryption function before the halves swap.

3. Initial & Final Permutations: The data undergoes an Initial Permutation before the 16 rounds begin, scrambling the bit order. After the final round, it undergoes the Final Permutation, which is the exact inverse of IP, to yield the final ciphertext.

4. The f Function Mechanics: Inside each round, the 32-bit right half is expanded to 48 bits via an Expansion Box (E-box) and XORed with a 48-bit round subkey. The result is fed into 8 Substitution Boxes (S-boxes), which compress the data back to 32 bits and provide crucial non-linearity.

5. P-Box & Subkey Generation: The 32-bit S-box output is shuffled by a Permutation Box (P-box) to ensure bit diffusion. Concurrently, the Key Schedule uses shifts and permutations to generate 16 distinct 48-bit subkeys from the master key, one for each round.

2. ***Discuss the feistel network properties used by DES***

1. Structural Reversibility: The most significant property is that the encryption and decryption processes are identical. Only the subkey schedule is reversed.

2. Non-Invertible Round Function : In a Feistel network, the internal round function does not need to be mathematically reversible because it is XORed with the data.

3. Product Cipher Execution: The Feistel structure facilitates the alternating application of Substitution (Confusion via S-boxes) and Permutation

4. Balanced Data Splitting: The network splits the 64-bit block into two equal 32-bit halves. In each round, only one half is processed while the other remains unchanged until they are XORed and swapped.

5. Iterative Resource Efficiency: The property of iteration means that a single round's logic is repeated 16 times.